

# ABDULSAMOD AZEEZ

Data Scientist | Machine Learning Engineer | Software Engineer | Python Engineer  
[LinkedIn](#) | [GitHub](#) | [Website](#)

## SUMMARY

Machine Learning Engineer with 3+ years of experience building and deploying AI-powered systems, including LLM applications and retrieval-augmented generation (RAG) pipelines. Passionate about leveraging data and AI to solve real-world challenges in emerging markets, particularly in healthcare, education, and digital access.

Experienced across the full ML lifecycle, from data engineering and modelling to scalable deployment using AWS, Docker, and Kubernetes. Strong background in NLP and transformer-based systems, with a proven ability to translate complex data into actionable insights. Mentor, researcher, and community contributor, having guided 50+ learners and supported the growth of data science ecosystems.

## TECHNICAL PROFICIENCIES

**Programming:** Python (FastAPI, Flask), JavaScript, Type Hinting, Static Analysis (MyPy)

**Databases:** MongoDB, PostgreSQL, MySQL, Redis, Qdrant

**AI/ML:** PyTorch, TensorFlow, Hugging Face Transformers, NLP, LLM Integration (Gemini, Claude), RAG, NVIDIA RAPIDS

**Data Science:** NumPy, Pandas, scikit-learn, Statistical Analysis

**Data Visualisation:** Matplotlib, Seaborn, Plotly, Tableau

**Test Automation:** Playwright, Appium, pytest, Selenium

**Cloud & DevOps:** AWS SageMaker, Docker, Kubernetes, Oracle Cloud Infrastructure (OCI), MLflow, BentoML, CI/CD, GitHub Actions

**Big Data & Streaming:** Apache Kafka, Redis (Pub/Sub, Streams), Real-time Data Pipelines

**Web Development:** FastAPI, Flask, Streamlit, RESTful APIs

**Tools & Platforms:** Git, GitHub, Linux, Jira

**Other:** Technical Writing, Research, Mentoring

## WORK EXPERIENCE

**Software Engineer** — **SimpleTest** (June 2024)

- Built a RAG pipeline (FastAPI, Gemini, Qdrant) that converts natural-language prompts into executable Salesforce test cases with streaming output and multi-tenant retrieval.
- Designed the test orchestration service around Kafka and Redis producer/consumer flows: producers dispatched jobs and result events, consumer groups distributed work across workers with at-least-once delivery, and MongoDB persisted the final state.
- Developed the browser and mobile automation engines (Playwright, Appium) with AI-generated locator evaluation, fallback strategies, MFA handling, and artefact capture to cloud storage.
- Built an autonomous browser agent using Claude and a ReAct loop to generate structured test cases from plain-English instructions.

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## ADDITIONAL EXPERIENCE

**Mentor** at **KaggleX BIPOC Mentorship Program**

(2023 - Present).

**Pioneer Campus Lead** at Data Science Nigeria, FULokoja Chapter

(2022 - 2024)

**Core Team Member and Mentor** at **Meta Developer Circles Lokoja**.

## EDUCATION

BSc, Computer Science

**Federal University Lokoja**

(Sep. 2017 - Jan. 2023)

## PUBLICATIONS

Performance and Accessibility Evaluation of University Websites in Nigeria  
([10.1109/ITED56637.2022.10051461](#)).

## CERTIFICATIONS

Kubernetes and Cloud Native Associate (KCNA)

The Linux Foundation – May, 2025.

Credential ID: LF-hij9qahoqh

## LANGUAGES

English.

## **Machine Learning Engineer — Turing (October 2025 - April 2026)**

- Built end-to-end ML pipelines (ingestion → training → evaluation → deployment → monitoring) on AWS, reducing model iteration time and standardising the path from prototype to production.
- Deployed and monitored LLM/ML services with Docker + Kubernetes; instrumented drift, latency, and accuracy metrics via MLflow and automated retraining on threshold breach.
- Optimised inference performance (batching, caching, quantisation) to meet strict latency SLOs in production.
- Translated ambiguous product requirements into reproducible, well-documented training workflows with versioned datasets and CI-tested model code.

## **Data Scientist — GOMYCODE (January 2023 - July 2025)**

- Mentored 50+ students across diverse backgrounds, improving access to data science education and enabling learners to apply analytical skills to real-world problems.
- Delivered project-based learning that emphasised practical application and deep understanding of data science concepts, tools, and methodologies.

## **Data Scientist and Machine Learning Engineer — Andela (January 2023 – May 2024)**

- Built and deployed scikit-learn / PyTorch models for classification and forecasting use cases, managing the full lifecycle from data preparation to production monitoring.
- Automated data pipelines with pandas and enforced data quality checks using Great Expectations, improving reliability and reducing manual effort.
- Deployed and scaled ML solutions in production with Docker, Kubernetes, and AWS, ensuring high availability and performance.
- Delivered model insights and recommendations to stakeholders through clear reporting, enabling data-driven business decisions.

## **PROJECTS**

### **AI-Powered Heart Disease Prediction System**

- Developed an end-to-end ML pipeline for early risk detection, handling data preprocessing, modelling, and deployment
- Achieved 92% accuracy and 0.91 AUC
- Deployed on AWS SageMaker with automated CI/CD pipelines
- Designed to support data-driven healthcare decision-making in resource-constrained environments

### **GenAI Chatbot with Retrieval-Augmented Generation (RAG)**

- Built an AI-powered chatbot using Hugging Face Transformers and vector search
- Implemented embedding-based retrieval to improve contextual understanding and response accuracy
- Developed backend APIs and interactive interfaces
- Improved task completion rates and user experience in conversational AI systems